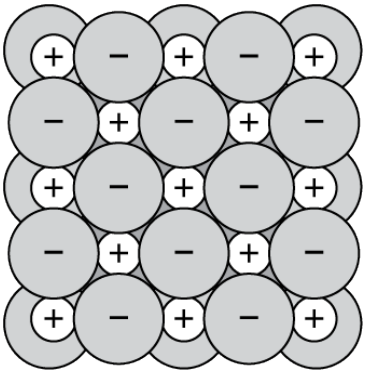
2.3

1.



Which of the following correctly indicates whether the solid represented by the particulate model shown above conducts electricity and explains why or why not?

(A) It conducts electricity because it is made of positive and negative ions.

(B) It conducts electricity because it is made of particles of different sizes.

(C) It does not conduct electricity because its ions cannot move freely within the solid. ✓

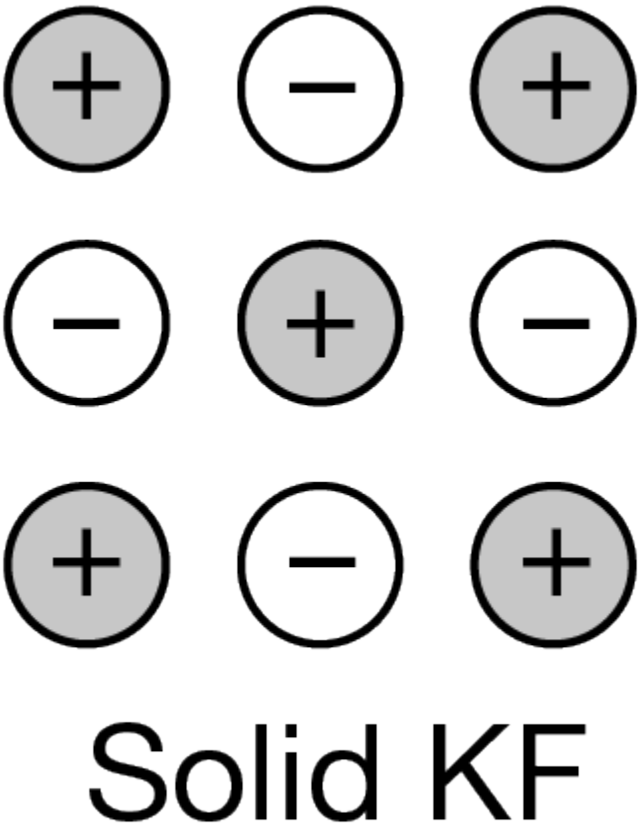
(D) It does not conduct electricity because there are small spaces between the particles.

Answer C

Correct. In the solid state, the ions are held by strong coulombic forces in fixed positions within the lattice; thus they cannot move or migrate through the lattice.

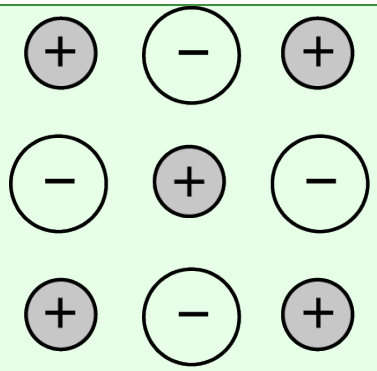
2.3

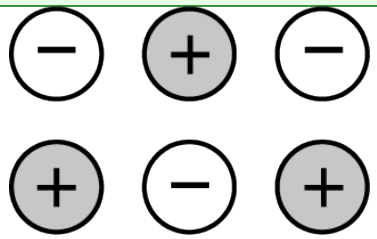
2.

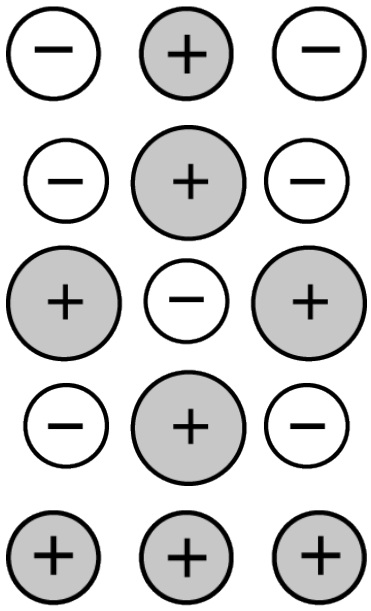


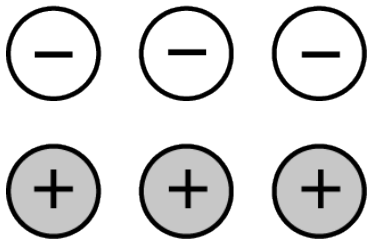
The particle-level diagram above represents the structure of solid KF. Although the molar mass of KCl is greater than that of KF, the density of KCl is actually less than that of KF. Which of the following representations of the structure of KCl best helps to explain this phenomenon?

2.3

(A) 

(B) 

(C) 

(D) 

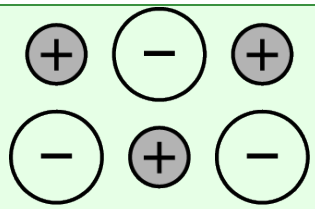
Answer A

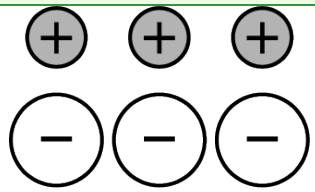
Correct. The diagram shows that the chloride ions are larger than the fluoride ions, which causes the

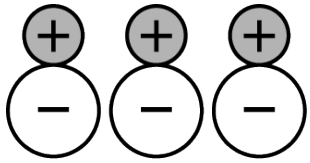
2.3

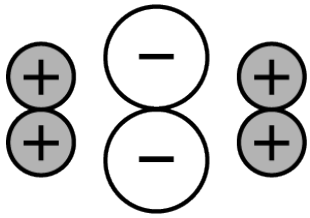
interionic distances in the solid to increase to the point where the effect of the larger spacing among ions in KCl is greater than the effect of the larger mass of the chlorine on the density.

3. The particles in solid KI, a stable ionic compound, are arranged to maximize coulombic attractions while minimizing coulombic repulsions among the particles. Which of the following diagrams best represents the structure of solid KI?

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. This diagram shows both types of charged ions being surrounded by ions of the opposite charge, thus maximizing attraction and minimizing repulsion.

2.3

4.

Compound	Lattice Energy $(\mathrm{kJ/mol})$
NaF	930
$\mathrm{MgF_2}$	2978

The energy required to dissociate an ionic solid into gaseous ions (lattice energy) for the compounds NaF and $\mathrm{MgF_2}$ is shown in the table above. On the basis of Coulomb's law, which of the following best helps to explain the large difference between the lattice energies of NaF and $\mathrm{MgF_2}$?

(A) The solubility of $\mathrm{MgF_2}$ is less than that of NaF.

(B) The electronegativity of Mg is greater than that of Na.

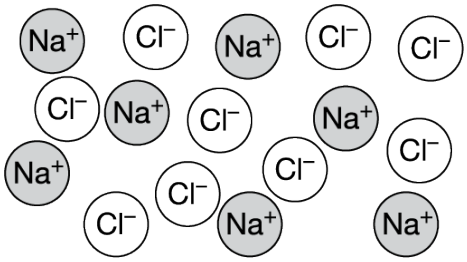
(C) The mass of the Mg cation is greater than that of the Na cation.

(D) The charge of the Mg cation is larger than that of the Na cation. ✓

Answer D

Correct. The coulombic attractions between the Mg cation and F anions are stronger because of the $+2$ charge on Mg as opposed to the $+1$ charge on Na. Stronger interionic attractions lead to a higher lattice energy.

5.



Molten (liquid) NaCl is represented by the particulate diagram shown above. Which of the following indicates whether $\mathrm{NaCl}(l)$ conducts electricity and best explains why or why not?

(A) It conducts electricity because Na is a metal.

(B) It conducts electricity because ions are free to move. ✓

(C) It does not conduct electricity because Cl is a nonmetal.

(D) It does not conduct electricity because there are no free electrons.

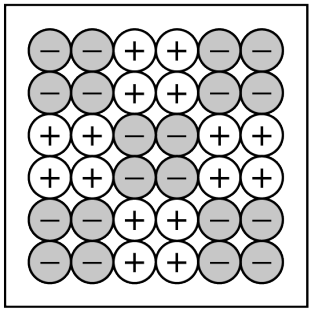
2.3

Answer B

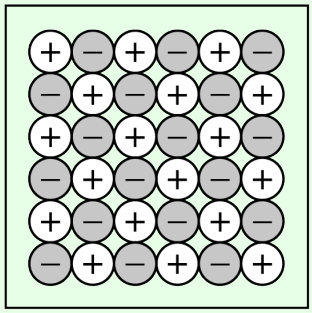
Correct. For a substance to conduct electricity, it must contain charged particles that are mobile. In molten salts such as $NaCl$, those charged particles are ions. (Metals conduct electricity via mobile electrons within a lattice of metal atoms.)

6. Of the following diagrams, which best represents the structure of solid KF ?

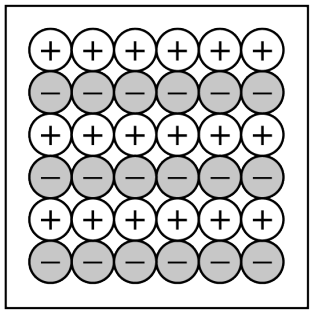
(A)



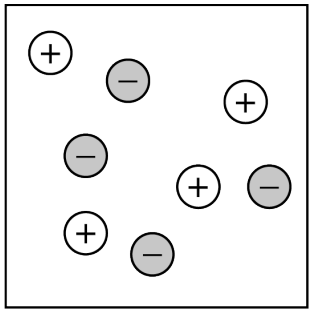
(B)



(C)



(D)



2.3

Answer B

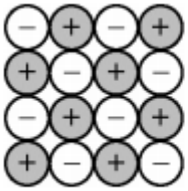
Correct. This diagram shows ions as being tightly packed surrounded by ions of the opposite charge, which is true in a crystal of an ionic compound such as KF. This arrangement represents the lowest energy state attainable by the constituent ions.

7. Which of the following diagrams best illustrates how a displacement in an ionic crystal results in cleavage and brittleness?

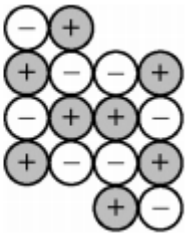
2.3

(A) ✓

Before Displacement

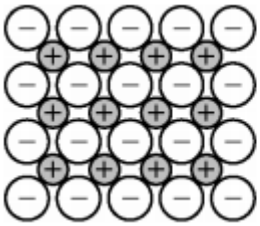


After Displacement

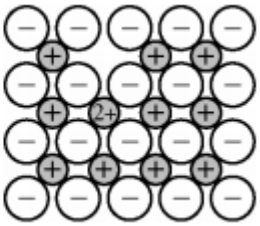


(B)

Before Displacement

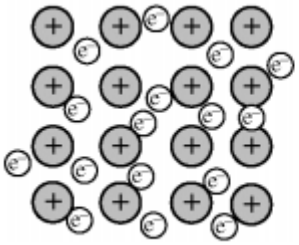


After Displacement

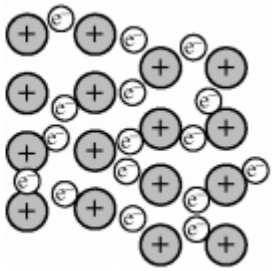


(C)

Before Displacement

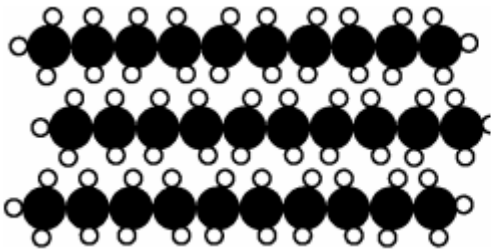


After Displacement

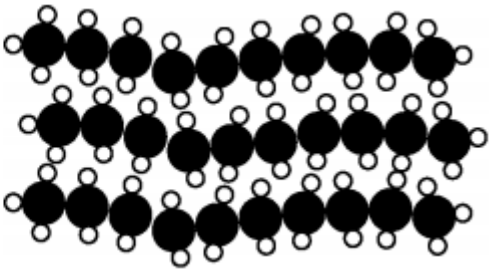


(D)

Before Displacement

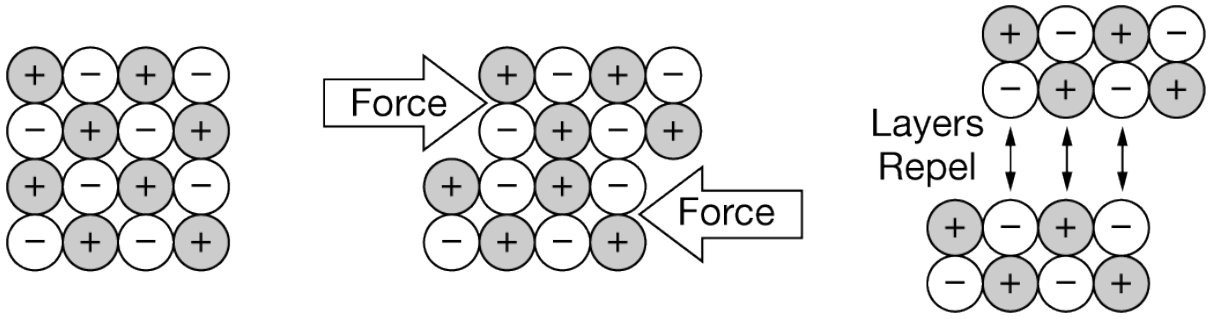


After Displacement



2.3

8.



The particulate-level diagram shown above best helps to explain which of the following properties of ionic solids?

(A) Density

(B) Brittleness ✓

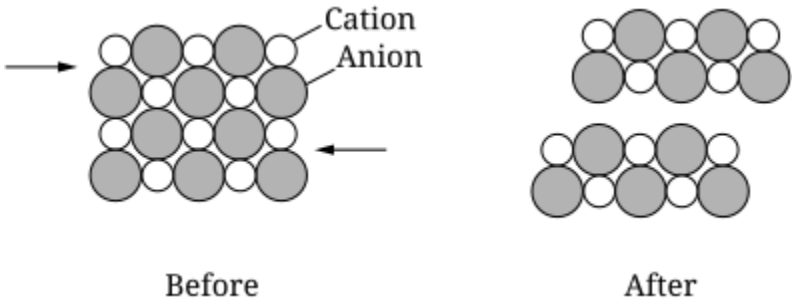
(C) Malleability

(D) Conductivity

Answer B

Correct. The brittleness of ionic solids is due to the internal repulsion of ion layers within the crystal lattice that occurs when the layers are displaced by an outside force parallel to the layers and they move relative to adjacent layers.

9.



When a stress is applied to an ionic crystal, the planes of the ions slide and separate due to the alignment of like charges, as shown in the diagram above. Which of the following characteristics of the crystal is represented by the observation?

(A) Luster

(B) Brittleness ✓

(C) High melting point

(D) Solubility in water

2.3

Answer B

Correct. Brittleness is the tendency of a material to break when pressure is applied.